Lab sessions – H9x34A
Electromagnetic and thermal analysis of a high voltage cable

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The objective of the lab sessions is to numerically characterise a power cable system using electric, magnetic and coupled magneto-thermal finite element (FE) models. Power cables play a key role in transmission and distribution networks. Their rating is limited by the electric stress in the insulation and, above all, by the maximum admissible conductor temperature, which depends on the losses in the conductors and in the metallic sheaths, armour and pipe. Numerical simulations help to design the cable system, predict ageing and degradation, and prevent failures. The lab follows the approach of [1], but you will work on a cable of your own choice, and verify and validate your results against analytical estimates and the cable datasheet.

By the end of the project, you should be able to: translate a real device and its datasheet into a well-posed, bounded 2D FE model and justify every simplifying assumption; implement and couple FE formulations in GetDP/Gmsh; verify (mesh convergence, analytical checks) and validate (datasheet) your results; use your model to propose and quantify design improvements; and present and defend your modelling choices orally.

Organisation and evaluation

The work is organised around three milestones. Milestones are *not* graded: they are meant to give you feedback in time. At each milestone you show your intermediate results during the in-class session.

Milestone	**Content**	**Date**
M1	Cable choice and electrodynamic analysis (Sections 1–2)	[November 10]
M2	Magnetoquasistatic analysis (Section 3)	[November 18]
M3	Coupled magneto-thermal analysis (Section 4)	[November 25]
Final	Code, slides and oral examination (Section 6)	[**December 1 or 2**]

The final evaluation is based on the submitted code, a slide deck, and a 15-minute presentation followed by a 15-minute Q&A (see Sections 6 and 7).

Starting files

As starting point, you get two FE formulations:

- electrodynamic formulation in `electrodynamic_formulation.pro`;
- magnetodynamic formulation in `magnetodynamic_formulation.pro`;
- coupled magneto-thermal formulation to be developed.

See GetDP tutorials for generiting the geometry and the profile to link the problem to the formulation. The file `Jacobian_Integration.pro` contains the Jacobian transformation and the numerical integration details (number of Gauss points), common definitions for all the formulations. The governing equations and the conventions to use are summarised in Appendix A; analytical reference formulas are given in Appendix B.

1 Cable selection and model set-up

You will start by choosing a power cable (high, medium voltage) from e.g. the following datasheets (single-core cables, three-core cables, magnetically shielded cables...):
NKT cables: 33 kV single core cable XLPE-AL
NKT cables: Medium Voltage Cables with XLPE Insulation
ABB cables: XLPE Submarine Cable Systems
Note: You may choose a single or three core cable but in any case, you are asked to model a **three-phase system** (three single-core cables in a given arrangement, or one three-core cable).

From the datasheets, you will need to get the dimensions, and the electromagnetic and thermal properties, similar to the example cable of Appendix C (Fig. 1, Tables 2 and 3). Datasheets rarely contain all material properties: values not in the datasheet must be taken from a cited source. The values of Appendix C are examples only and must be justified if you use them. You will also need to place the cable in an environment (air, ground, water,...) with the corresponding properties, and bound the problem.

Q1.1 Describe your cable: construction, dimensions, materials, rated voltage, rated current (ampacity) and the *laying conditions* for which that rating is given (burial depth, soil thermal resistivity, ambient temperature, arrangement). Give a table of electromagnetic and thermal properties with a source for each value.

Q1.2 Choose the environment consistently with the datasheet rating conditions, and justify its properties.

Q1.3 State the **sheath bonding** you assume: single-point bonded (no net sheath current) or solidly bonded (circulating sheath currents allowed), and how the armour and pipe are connected. This choice strongly affects the losses in Section 3.

Q1.4 State the excitation you impose: phase-to-ground voltage and phase current, as peak phasors, with the phase sequence (see the conventions in Appendix A).

2 Electrodynamic analysis

The goal of this analysis is to determine the electric stress within the cable. Too high electric field in the cable can cause partial discharges, dielectric damage or early ageing. The problem can be solved by using an electrodynamic model accounting for the resistive and capacitive effects, see (6) in Appendix A. Defects (voids, bubbles, water trees) in the XLPE insulation may cause stronger electric fields, see e.g. [1].

Q2.1 Define your computational domain, the equations solved and the boundary conditions. Where do you truncate the domain, and why? Discuss the role of the grounded screens: which regions actually influence the field in the insulation?

Q2.2 Can you simplify the geometry of your cable without degrading the precision (e.g. disregard details in the geometrical description)? Quantify the effect of each simplification.

Q2.3 How do the results vary with the mesh refinement? Plot the maximum field in the insulation and the capacitance versus the number of degrees of freedom, and identify where local refinement is needed.

Q2.4 Define the admissible field strength at the conductor screen and at the core screen from the datasheet (or, if absent, from a cited reference). Is the healthy cable within these limits?

Q2.5 Add a defect to the insulation of your cable. State its nature (e.g. gas void, water tree, protrusion of the semiconducting screen), its properties, size and position, and discuss what a defect in a 2D cross-section represents in 3D. Compare the field with and without defect along a radial cut. Does the cable withstand the defect?

Q2.6 Compute the resistive and displacement current densities in the cable. Is current conservation satisfied? Which current dominates in each material, and why?

Q2.7 Compute the per-unit-length capacitance (μF/km) and compare with the analytical value (13) and with the datasheet. State the assumptions of the analytical formula (in particular, how the semiconducting layers are treated). Does the capacitance vary with frequency? Why (not)?

Q2.8 Propose an improvement of the cable design with respect to the electric stress, and support it quantitatively.

3 Magnetoquasistatic analysis

Let us add now the magnetic part. We solve the time-harmonic magnetoquasistatic problem via a magnetic vector potential formulation, see (8) in Appendix A.

Q3.1 Define your computational domain, the equations solved and the boundary conditions. Where do you truncate the domain, and why? How do you impose the currents in the phase conductors?

Q3.2 Conductor model. Compare the skin depth in the conductor with its radius. Model the phase conductors (a) as stranded conductors (imposed uniform current density) and (b) as massive conductors (imposed net current, eddy currents allowed). Which assumption is more realistic for your cable, and how does it affect the results?

Q3.3 Bonding. Implement the bonding assumed in Q1.3 with global quantities (net current and voltage per unit length in the sheaths, armour and pipe). Explain why using only $\underline{j} = -\imath\omega\sigma\underline{a}$ in these regions does not correspond to a well-defined bonding condition.

Q3.4 Can you simplify the geometry without degrading the precision? Are the simplifications of Section 2 still valid here?

Q3.5 Check the mesh quality and refine where needed (element size compared with the skin depth in each conducting region). How do losses, resistance and inductance vary with the mesh refinement?

Q3.6 Get the magnetic flux density (or induction) in and around your cable.

Q3.7 Get the current density in the conducting parts of your cable.

Q3.8 Get the time-averaged Joule losses (kW/km) in each conducting part (conductors, sheaths, armour, pipe, environment).

Q3.9 Get the per-unit-length AC resistance (Ω/km), defined from the total losses. Compare it with the DC resistance and with the datasheet. Separate the contributions of skin/proximity effect and of sheath/armour losses.

Q3.10 Get the per-unit-length inductance (mH/km). How do you define it for a three-phase system? Does it vary with frequency? Compare with the datasheet.

Q3.11 Propose a design change that reduces the losses, and support it quantitatively.

4 Coupled magneto-thermal analysis

The time-averaged Joule losses of the magnetoquasistatic model are the heat source of the steady-state heat conduction problem (10)[1].

[1]The thermal source is computed from the complex MVP {a}, without `Dof`. Notation: `<a>[ my_function ]` must appear in front of any function applied to {a} in the thermal formulation to indicate that the result of the operation is a real quantity.

Linear case (constant conductivity)

Q4.1 Define the equations solved. How are the complex magnetoquasistatic fields turned into a real heat source?

Q4.2 Define the thermal computational domain and boundary conditions. How is the cable cooled? Apply a **convective** condition (11) at the ground (or water/air) surface with a justified heat transfer coefficient. Where do you truncate the domain laterally and in depth? Why must the thermal domain be much larger than the electromagnetic one? Do you use one mesh or two?

Q4.3 The ambient temperature must be consistent with the datasheet rating conditions. How do the results vary with the mesh refinement and with the domain size?

Q4.4 Get the temperature distribution in the cable and identify the hot spot. Is it within the operating temperature limits (typically 90°C for XLPE in continuous operation; check the datasheet)?

Q4.5 Get the temperature distribution around the cable. What is the maximum temperature at the ground/air interface? Compare with a Dirichlet condition $T = T_{\text{amb}}$ at the surface.

Q4.6 **Validation.** Compare your conductor temperature with the analytical estimate of Appendix B (IEC 60287 approach). At the datasheet ampacity and laying conditions, does your model predict the admissible conductor temperature? Explain the differences.

Nonlinear case (temperature-dependent conductivity)

Till this point, we have considered that the conductivity σ does not depend on the temperature T. Let us now account for the temperature-dependent electrical conductivity (12) in the copper and aluminum parts. Solve the coupled nonlinear problem, iterating between the magnetoquasistatic and the thermal problems.

Q4.7 Define your convergence criterion precisely: which quantity, which norm, which tolerance (e.g. relative increment of the temperature below 10^{-6}). How many nonlinear iterations do you need? Is the convergence influenced by the mesh density? Plot the convergence history.

Q4.8 Compare all magnetoquasistatic and thermal quantities of Sections 3 and 4 (flux density, current density, losses, AC resistance, inductance, temperatures) with the linear case, with field maps. Which quantities change significantly, and why?

Q4.9 Is the linear model conservative or not? What does this imply for the cable rating?

5 Additional investigation (not optional)

Depending on the type of cable system you choose, it may be useful to adapt e.g.:

- An under-sea cable may be laying in a pipe, buried in soil or directly in water: what would be the impact of the different environment?
- Different cooling conditions: vary the burial depth, include a duct, consider soil drying around the cable, or the distance between the cable and the ground for an aerial cable...
- Single-point versus solid bonding, or cross-bonding.
- Arrangement of single-core cables (trefoil versus flat), spacing, several circuits, or multiple cables on a cable tray: what is the best cable arrangement for a given specification?
- Nonlinear magnetic behaviour of the steel armour.
- Symmetry or (anti-)periodicity conditions to reduce the model.
- Monolithic coupling of the magnetoquasistatic and thermal problems.

- A 2D axisymmetric model of a spherical void in the insulation.
- Any idea you may get when working on this cable project!

Once you have presented milestone M1, the teaching team will indicate the precise additional investigation to perform. It could be one of the possibilities above or something similar.

6 Deliverables

Code. The `geo` and `pro` files you have created and/or modified should be included. The submitted version is identified by the git tag `final`. The repository must contain a `README` stating the Gmsh and GetDP versions used, the exact commands (or Onelab parameter settings) that reproduce each result in the slides, and what each file does. All cable, material and excitation data must be exposed as Onelab parameters. The code must run without errors and reproduce the results shown in the slides.

Slides. Submit the slides (PDF) before the oral examination. They consist of

- a **main part** of about 15 slides for the 15 minute talk, presenting your modelling approach, verification, validation, main results and design proposals;
- an **appendix** with at least one slide per question (Q1.1 to the last question of Section 4), labelled with the question number, and including the field maps listed below;
- the completed **results summary table** (Table 1).

Electrodynamic analysis:

- Post-processing map of the electric potential in the cable area.
- Post-processing map of the electric field in the cable area (with and without defect).
- Post-processing map of the displacement current in the cable area (with and without defect).
- Electric field stress of the cable in a phase (cut in the radial direction) with and without defect in the insulation.
- Per-unit-length capacitance (μF/km), comparison with analytical capacitance and datasheet.

Magnetoquasistatic analysis:

- Post-processing map of the magnetic flux density norm in the cable area.
- Post-processing map of the current density norm in the conducting parts of your cable.
- Joule losses (kW/km) in each conducting part of your cable.
- Per-unit-length AC resistance (Ω/km).
- Per-unit-length inductance (mH/km).

Coupled magneto-thermal analysis:

- Post-processing map of the temperature in the cable.
- Post-processing map of the temperature outside the cable.
- Convergence history of the nonlinear iterations.

For the nonlinear coupled case with $\sigma(T)$, the same maps and values, compared with the linear case.

Quantity	FE (linear)	FE ($\sigma(T)$)	Reference*
Max. field at conductor screen (kV/mm)		–	
Max. field with defect (kV/mm)		–	
Capacitance (μF/km)		–	
DC resistance (Ω/km)		–	
AC resistance (Ω/km)			
Inductance (mH/km)			
Losses (kW/km): conductors / sheaths / armour and pipe			
Max. conductor temperature (°C)			
Max. surface temperature (°C)			
Nonlinear iterations	–		–

Table 1: Results summary table (to be completed in the slides). *Analytical value (Appendix B) or datasheet; state which.

Oral examination. Each student presents for 15 minutes, followed by a 15-minute Q&A. Questions may address any item of this assignment, the code, and the assumptions and limitations of your model. You must be able to explain every line of code you submit.

Use of AI tools. You may use AI tools for programming support, provided you declare how you used them in the `README`. You remain fully responsible for, and will be questioned on, all submitted code and results.

7 Assessment

Assesment is based on the submitted code, the slides, the presentation and the Q&A. The presentation must include: Modelling choices, Verification and validation, Results, and Design and additional investigation. Each criterion has a weight in the final grade as shown in the table below. Pay attention to the time managment of your presentation, it must not exceed 15 minutes.

Criterion	Description	Weight
Modelling choices	Well-posed domains, boundary conditions, excitation, bonding and simplifications, all justified.	20 %
Verification and validation	Mesh convergence, analytical checks, comparison with the datasheet, critical discussion of discrepancies.	20 %
Results	Completeness and correctness of the requested quantities and field maps; reproducible code.	20 %
Design and additional investigation	Quantified design proposals; depth and originality of the additional investigation.	15 %
Q&A	Understanding of the physics, the formulations and the code.	25 %

A Governing equations and conventions

Conventions. All fields are time-harmonic phasors at angular frequency $\omega = 2\pi f$, with $x(t) = \mathrm{Re}\{\underline{x}\, e^{\imath\omega t}\}$, i.e. phasors are **peak** amplitudes, as in GetDP. Report voltages and currents as RMS values and losses as time averages. For a balanced system with RMS line voltage U and RMS phase current I, the phasors imposed on phase $k = 0, 1, 2$ are

$$\underline{V}_k = \frac{\sqrt{2}\,U}{\sqrt{3}}\, e^{-\imath 2\pi k/3}\,, \qquad \underline{I}_k = \sqrt{2}\, I\, e^{-\imath 2\pi k/3}\,. \tag{1}$$

For example, $U = 132\,\mathrm{kV}$ gives a phase-to-ground peak voltage of about $107.8\,\mathrm{kV}$.

Electrodynamics. The time-harmonic electrodynamic problem reads

$$\mathrm{curl}\,\underline{e} = 0\,, \tag{2}$$

$$\mathrm{curl}\,\underline{h} = \underline{j} + \imath\omega\underline{d} \Rightarrow \mathrm{div}\,(\underline{j} + \imath\omega\underline{d}) = 0\,, \tag{3}$$

$$\underline{j} = \sigma\,\underline{e}\,, \tag{4}$$

$$\underline{d} = \epsilon\,\underline{e}\,, \tag{5}$$

with $\underline{e}$ the electric field, $\underline{h}$ the magnetic field, $\underline{j}$ the electric current density, $\underline{d}$ the electric flux density, σ the conductivity and ϵ the permittivity. With the electric scalar potential v, $\underline{e} = -\mathrm{grad}\, v$, one gets

$$-\mathrm{div}\left((\sigma + \imath\omega\epsilon)\,\mathrm{grad}\, v\right) = 0\,, \tag{6}$$

completed with Dirichlet conditions: $v = \underline{V}_k$ on conductor k and $v = 0$ on the grounded screens.

Magnetoquasistatics. In 2D, with $\underline{a} = a_z\,\hat{z}$ and $\underline{b} = \mathrm{curl}\,\underline{a}$:

$$\mathrm{curl}\,(\nu\mathrm{curl}\,\underline{a}) - \underline{j} = 0\,, \tag{7}$$

where

$$\underline{j} = \begin{cases} \underline{j}_s = \dfrac{\underline{I}_k}{\mathcal{A}_k}\,\hat{z} & \text{stranded conductor } k, \\ -\sigma\,(\imath\omega\underline{a} + \mathrm{grad}\, v) & \text{massive conductor or sheath,} \end{cases} \tag{8}$$

with $\mathcal{A}_k$ the cross-section of conductor k, and $\mathrm{grad}\, v = -(\underline{U}_m/\ell)\,\hat{z}$ uniform in each massive region m. Each massive region is closed either by an imposed net current $\int_{\Omega_m} \underline{j}\cdot\hat{z}\,\mathrm{d}s = \underline{I}_m$ (e.g. $\underline{I}_m = 0$ for a single-point bonded sheath) or by an imposed voltage per unit length (e.g. $\underline{U}_m = 0$ for solid bonding). At the boundary of the computational domain, $\hat{n}\times\underline{a} = 0$. The displacement-current term $-\omega^2\epsilon\underline{a}$ of `darwin_formulation.pro` is negligible at power frequency.

The time-averaged Joule loss density and the AC resistance per phase are

$$p = \frac{|\underline{j}|^2}{2\sigma}\,, \qquad R_{\mathrm{AC}} = \frac{P_{\mathrm{tot}}}{3I^2}\,, \tag{9}$$

with P_{tot} the total losses per unit length and I the RMS phase current. Note that p must be computed with the *total* current density, not only with $-\imath\omega\sigma\underline{a}$.

Thermal. The steady-state heat conduction equation reads

$$-\mathrm{div}\,(\kappa\,\mathrm{grad}\, T) = Q\,, \tag{10}$$

with T the temperature (K), κ the thermal conductivity (W/(m·K)), and $Q = p$ the heat source (W/m^3) given by (9). At a surface in contact with air or water, a convective condition

$$-\kappa\,\frac{\partial T}{\partial n} = h\,(T - T_{\mathrm{amb}}) \tag{11}$$

applies, with h the heat transfer coefficient (W/(m^2·K)). The conditions on the truncation boundaries (fixed temperature or zero flux) must be justified.

Temperature-dependent conductivity.

$$\sigma(T) = \frac{\sigma_0}{1 + \alpha(T - T_{\mathrm{ref}})}\,, \tag{12}$$

with σ_0 the conductivity at $T_{\mathrm{ref}} = 20°\mathrm{C}$, and α equal to $0.00386\,\mathrm{K}^{-1}$ for the copper and $0.00390\,\mathrm{K}^{-1}$ for the aluminum.

B Analytical reference values

Capacitance of a single screened core with insulation between radii r_i and r_o (to be defined with respect to the semiconducting layers):

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln(r_o/r_i)}\,. \tag{13}$$

DC resistance and skin depth:

$$R_{\mathrm{DC}} = \frac{1}{\sigma \mathcal{A}}\,, \qquad \delta = \sqrt{\frac{2}{\omega \mu \sigma}}\,. \tag{14}$$

Conductor temperature (IEC 60287 approach, simplified). For a buried cable, the conductor temperature rise can be estimated from a ladder of thermal resistances per unit length (T_1: insulation, T_2: bedding, T_3: serving, T_4: soil), with the losses in the conductors (W_c), sheaths ($\lambda_1 W_c$) and armour ($\lambda_2 W_c$). The soil thermal resistance of a single cable of outer diameter D_e buried at depth L is given by Kennelly's formula:

$$T_4 = \frac{\rho_T}{2\pi} \ln\left(u + \sqrt{u^2 - 1}\right), \qquad u = \frac{2L}{D_e}\,, \tag{15}$$

with $\rho_T = 1/\kappa$ the soil thermal resistivity. This formula assumes an isothermal ground surface. See IEC 60287-1-1 [2] and IEC 60287-2-1 [3] for three-core cables, sheath and armour loss factors, and groups of cables.

C Example cable

The cable in Fig. 1 is a three-core cable with stranded copper conductors and cross linked polyethylene (XLPE) as insulating material. A semi-conductor layer surrounds the copper and the XLPE insulation. Each single core is surrounded by an aluminum polyethylene laminated (APL) sheath. The group of the three cores is sheathed by polyethylene. Both wire armour and external pipe are made of steel. Finally, the external covering of the pipe is a polyethylene layer. The cable is placed at a depth of 1.2 m. The three-phase voltages and currents are balanced at frequency $f = 50\,\mathrm{Hz}$. The RMS value of the line voltage is 132 kV. According to the cable technical data, the maximum field strength at the conductor and core screens is 10.3 kV/mm and 4.9 kV/mm, respectively. The ambient temperature is $T_{\mathrm{amb}} = 15°\mathrm{C}$, and the operating temperature of the cable varies between 70°C and 90°C [1].

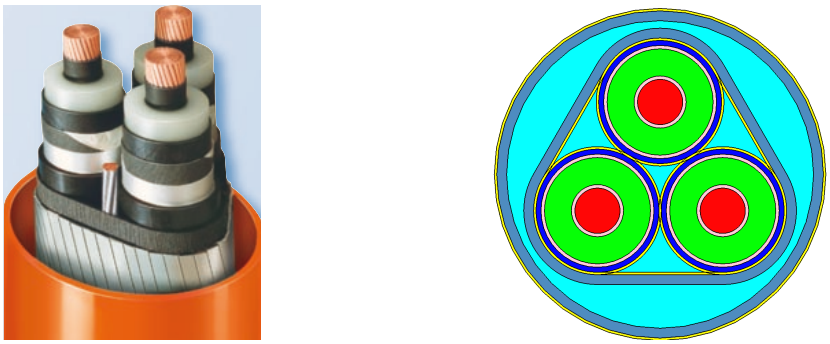


Figure 1: Left: Picture of a high voltage three-phase underground cable from NKTcables. Right: Geometrical model, the HV cable comprises: three copper stranded conductors (red), XLPE (green), semiconductors (pink and yellow), APL-sheath (dark blue), steel armour and pipe (grey), air (light blue).

Parameter	**dimensions [mm]**
Conductor diameter	18.4
Inner semiconductor thickness	1.4
XLPE-insulation thickness	11.0
Outer semiconductor thickness	1.4
APL-sheath thickness	2.0
Polyethylene-sheath thickness	1.0
Steel-armour thickness	2.0
Steel-pipe thickness	4.0
Polyethylene-covering thickness	1.0
Outer diameter	135.0

Table 2: Example geometrical data.

Material	ϵ_r	μ_r	σ [S/m]	κ [W/(m·K)]
copper	1	1	5.99e7	400
steel	1	4[a]	4.7e6	50.2
aluminum	1	1	3.77e7	237
polyethylene	2.25	1	1.0e-18	0.46
semiconductor	2.25	1	2	10
XLPE	2.5	1	1.0e-18	0.46
soil[b]	10	1	1.0e-2	1.0[c]

Table 3: Example electromagnetic and thermal material properties.
[a]Value from [1]. The permeability of armour steel depends strongly on the grade and field level (often in the hundreds, and nonlinear); justify your choice and discuss its influence on the losses. [b]Indicative values; soil properties vary widely with composition and moisture content. [c]Corresponds to the thermal resistivity of 1 K·m/W commonly used for cable rating; about 0.4 W/(m·K) for dry soil.

References

[1] S. Conti, E. Dilettoso, S.A. Rizzo, "Electromagnetic and thermal analysis of high voltage three-phase underground cables using finite element method", *In IEEE International Conference on Environment and Electrical Engineering and 2018 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, Palermo, Italy, June 12-15, 2018. https://ieeexplore.ieee.org/document/8525354.

[2] IEC 60287-1-1, *Electric cables – Calculation of the current rating – Part 1-1: Current rating equations (100 % load factor) and calculation of losses – General.*

[3] IEC 60287-2-1, *Electric cables – Calculation of the current rating – Part 2-1: Thermal resistance – Calculation of thermal resistance.*